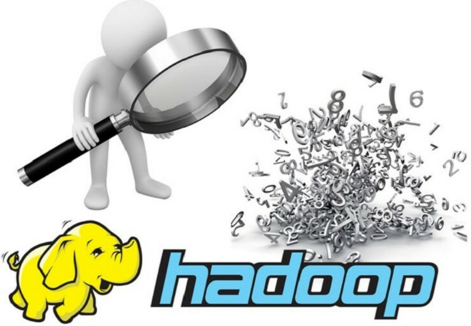
Use Hadoop to Handle Big Data



Big Data now means big business. The world is constantly accumulating volumes of raw data in various forms such as text, MP3 or Jpeg files, which need to be processed, if any value can be derived from them. Apache Hadoop is open source software that can handle Big Data. So here's a chance to learn how to install Hadoop and play around with it.

Big Data is currently making waves across the tech field. Everyone knows that the volume of data is growing day by day. Old technology is unable to store and retrieve huge amounts of data sets. With a rapid increase in the number of mobile phones, CCTVs and the usage of social networks, the amount of data being accumulated is growing exponentially. But why is this data needed? The answer to this is that companies like Google, Amazon and eBay track their logs so that ads and products can be recommended to customers by analysing user trends. According to some statistics, the New York Stock Exchange generates about one terabyte of new trade data per day. Facebook hosts approximately 10 billion photos, taking up one petabyte of storage. The Big Data we want to deal with is of the order of petabytes— 10^{12} times the size of ordinary files. With such a huge amount of unstructured data, retrieval and analysis of it using old technology becomes a bottleneck.

Big Data is defined by the three Vs—volume, velocity and variety. We're currently seeing exponential growth in data storage since it is now much more than just text. This large volume, indeed, is what represents Big Data. With the rapid increase in the number of social media users, the speed at which data from mobiles, logs and cameras is generated is what the second 'v'(for velocity) is all about. Last of all, variety represents different types of data. Today data is in different formats like text, mp3, audio, video, binary and logs. This data is unstructured and not stored in relational databases.

In order to solve the problem of data storage and fast retrieval, data scientists have burnt the midnight oil to come up with a solution called Hadoop. It was created by Doug Cutting and Mike Cafarella in 2005. Cutting, who was working at Yahoo at that time, named this solution after his son's toy elephant.

So what is Hadoop? It is an open source framework that allows the storage and processing of Big Data in a distributed environment across clusters of computers using simple programming models. It is designed to scale up from single servers to thousands of machines, each offering local computation and storage. The core of Apache Hadoop consists of the storage part (Hadoop distributed file system) and its processing part (MapReduce). Hadoop splits files into large blocks and distributes them amongst the nodes in the cluster. It should be noted that Hadoop is not OLAP (online analytical processing) but batch/offline oriented.

The challenge with Big Data is whether the data should be stored in one machine. Hard drives are approximately 500GB in size. Even if you add external hard drives, you can't store the data in petabytes. Let's say you add external hard drives and store this data, you wouldn't be able to open or process